

LAFAYETTE, LA, USA

WMO: 722405

Lat: 30.205N

Lon: 91.988W

Elev: 38

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.9	(c) 33.5	(d) 16.2	(e) 12.5	(f) 35.9	(g) 21.2	(h) 15.9	(i) 40.8	(j) 23.2	(k) 57.3	(l) 20.9	(m) 56.7	(n) 7.3	(o) 10	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 16.3	(c) 94.9	(d) 77.7	(e) 93.2	(f) 77.6	(g) 91.6	(h) 77.3	(i) 80.8	(j) 88.6	(k) 80.1	(l) 88.0	(m) 79.4	(n) 87.3	(o) 7.0	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.7	83.5	78.1	146.1	83.2	77.3	142.1	82.8	44.2	88.7	43.4	87.9	42.7	87.2	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.1	(b) 18.0	(c) 16.2	(e) 24.5	(f) 97.8	(g) 4.2	(h) 2.4	(i) 21.5	(j) 99.5	(k) 19.0	(l) 100.9	(m) 16.7	(n) 102.3	(o) 13.6	(p) 104.1		
(a) 20.1	(b) 18.0	(c) 16.2	(e) 22.1	(f) 83.0	(g) 3.8	(h) 2.4	(i) 19.3	(j) 84.7	(k) 17.1	(l) 86.1	(m) 15.0	(n) 87.5	(o) 12.3	(p) 89.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	52.8	56.6	62.6	69.1	76.4	81.3	82.9	83.1	79.2	70.5	61.0	54.8
	DBStd	13.01	10.22	9.62	8.36	6.60	4.87	3.14	2.64	2.97	4.44	7.22	8.41	9.81
	HDD50	216	91	41	11	0	0	0	0	0	0	0	12	61
	HDD65	1388	395	262	145	36	2	0	0	0	1	33	172	342
	CDD50	7247	176	226	400	574	820	940	1021	1027	877	635	342	209
	CDD65	2945	16	27	69	160	357	490	556	562	427	204	52	25
	CDH74	27416	26	69	303	1106	3129	4965	5949	6015	3898	1590	288	78
	CDH80	10768	0	1	19	202	1101	2132	2552	2700	1570	460	30	1
Wind	WSAvg	6.6	8.1	8.4	8.1	7.9	6.8	5.6	4.9	4.8	5.3	5.8	6.5	7.4
Precipitation	PrecAvg	61.5	5.6	4.4	3.9	5.0	5.0	6.8	6.0	5.7	5.3	4.7	4.3	4.8
	PrecMax	85.3	13.3	9.1	10.6	11.9	13.5	17.9	9.3	27.9	10.3	13.9	13.8	13.5
	PrecMin	40.4	0.8	0.7	0.3	0.5	0.3	0.9	1.8	1.5	2.0	0.2	1.2	2.3
	PrecStd	11.3	3.6	2.4	2.4	2.9	3.4	3.7	1.9	4.8	2.4	3.9	3.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.9	79.0	82.2	87.2	92.8	96.7	96.1	97.9	95.3	90.4	83.2	79.4
		MCWB	68.3	70.2	69.5	71.1	74.6	76.5	78.1	77.7	76.3	74.2	71.4	70.9
	2%	DB	73.9	76.2	79.9	84.4	90.0	93.4	93.9	95.0	92.2	87.5	80.4	76.3
		MCWB	66.9	68.3	68.5	71.3	74.1	76.6	78.2	78.2	76.0	74.3	70.4	69.5
	5%	DB	71.6	73.7	77.5	82.3	87.9	91.4	92.3	93.2	90.4	85.0	77.4	73.3
		MCWB	65.9	67.3	67.8	70.1	74.0	76.8	78.2	78.3	75.8	72.5	68.2	68.2
	10%	DB	68.7	71.5	75.1	80.3	86.1	89.7	90.6	91.3	88.4	82.3	74.7	70.5
		MCWB	64.0	65.9	66.9	69.0	73.4	76.5	78.0	78.0	75.3	71.4	67.2	66.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.2	72.7	73.1	76.1	79.3	81.5	81.5	82.0	80.5	79.0	75.2	73.3
		MCDB	74.4	75.8	78.0	82.1	86.3	88.1	89.6	90.4	87.3	83.9	79.3	76.8
	2%	WB	69.3	70.9	71.4	74.3	77.5	80.3	80.5	80.6	79.3	77.4	73.1	71.3
		MCDB	72.5	74.4	75.8	79.8	84.5	87.3	88.6	89.1	86.1	82.9	76.9	74.4
	5%	WB	67.2	69.2	70.4	73.1	76.4	79.3	79.9	80.0	78.3	76.0	71.2	69.4
		MCDB	70.0	72.3	74.9	78.5	83.3	86.6	88.1	88.5	85.2	81.7	75.4	72.9
	10%	WB	64.8	67.2	69.0	72.0	75.3	78.3	79.1	79.2	77.4	74.5	69.0	66.7
		MCDB	68.0	70.4	73.6	77.4	82.3	85.5	87.1	87.6	84.4	79.7	73.7	69.9
Mean Daily Temperature Range	5% DB	MDBR	18.0	17.6	18.5	18.8	17.6	16.1	15.3	16.3	17.4	20.0	20.2	18.3
		MCDBR	17.7	17.2	17.7	18.2	18.3	18.0	17.0	18.2	18.9	19.8	20.4	18.4
		MCWBR	12.1	10.8	9.1	8.0	6.3	5.6	5.5	5.7	6.5	8.4	11.6	12.7
	5% WB	MCDBR	15.7	15.0	15.1	15.1	15.9	15.4	15.5	16.3	15.6	16.2	17.7	16.8
		MCWBR	13.1	11.5	10.2	8.6	6.3	6.3	6.0	6.1	6.4	8.6	12.8	13.5
Clear-Sky Solar Irradiance	taub	0.337	0.354	0.378	0.420	0.447	0.475	0.494	0.487	0.446	0.380	0.357	0.350	
	taud	2.506	2.466	2.387	2.276	2.235	2.201	2.183	2.200	2.283	2.444	2.455	2.473	
	Ebn at Noon	284	289	287	278	269	260	255	256	263	276	274	272	
	Edh at Noon	28	32	37	43	45	46	47	46	41	32	30	28	
All-Sky Solar Radiation	RadAvg	920	1073	1466	1789	1968	1925	1848	1748	1604	1415	1066	822	
	RadStd	90	160	129	131	117	161	138	157	142	166	102	85	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+292	+189
Regional (26 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+214	+146

Nomenclature: See separate page